FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2016**

PART TEST – IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+2 marks** for correct answer. There is no negative marking.

Section-A (11 to 15) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

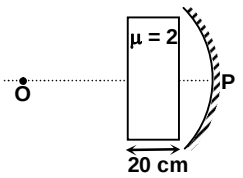
SECTION – A

Straight Objective Type

This section contains **10 multiple** choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which only ***ONE*** is correct.

1. Two radioactive materials X_1 and X_2 have decay constants 10 λ and λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be 1/e after a time
   (A) 1/(10 λ) (B) 1/(11λ)
   (C) 11/(10 λ) (D) 1/ (9λ)

2. Radius of curvature of the concave mirror is 60 cm. In front of this a glass slab is situated very close to the mirror as shown in the figure. A point object O is lie at a distance of 100 cm from the pole then the position of the image from the pole is
   (A) 45 cm (B) 55 cm
   (C) 40 cm (D) 60 cm



3. The minimum distance between an object and its real image formed by a thin convex lens of focal length 30 cm is
   (A) 45 cm (B) 50 cm
   (C) 75 cm (D) 120 cm

4. A coil of resistance R and inductance L is connected across an a.c. power supply of r.m.s. voltage V. The power dissipated in the coil is,
   (A) V^2/R (B) $\frac{V^2}{\sqrt{(R^2+\omega^2L^2)}}$
   (C) $\frac{V^2R}{(R^2+\omega L^2)}$ (D) zero

5. In young's double slit experiment, the intensity of light at a point on the screen where the path difference is λ is k units; λ being the wavelength of light used. The intensity at a point where the path difference is $\lambda/4$ will be
   (A) k/4 (B) k/2
   (C) k (D) zero

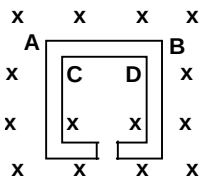
Space for rough work

6. In Young's double slit experiment the two slits act as coherent sources of equal amplitude A and of wavelength λ. In another experiment with same set up the two slits are sources of equal amplitude A and wavelength λ, but are incoherent. The ratio of intensity of light at the mid point of the screen in the first case to that in the second case is
(A) 1 : 1 (B) 1 : 2
(C) 2 : 1 (D) 4 : 1

7. The wavelength of K_α line for an element of atomic number 29 is λ. Then the wavelength of K_α line for an element of atomic no 15 is (Take Mosley's constant b = 1 for both elements)
(A) $\frac{29}{15}\lambda$ (B) $\frac{28}{15}\lambda$
(C) 4λ (D) 2λ

8. A certain radioactive material can undergo three different types of decay, each with a different decay constant λ_1, λ_2 and λ_3. Then the effective decay constant is
(A) $\lambda_{eff} = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3}$ (B) $\frac{1}{\lambda_{eff}} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$
(C) $\lambda_{eff} = \lambda_1 + \lambda_2 + \lambda_3$ (D) $\frac{1}{\lambda_{eff}} = \frac{1}{3}\left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3}\right)$

9. A wire is bent to form the double loop shown in the figure. There is a uniform magnetic field directed into the plane of the loop. If the magnitude of this field is decreasing, current will flow from
(A) A to B and C to D (B) B to A and D to C
(C) A to B and D to C (D) B to A and C to D



10. In the ideal double-slit experiment, when a glass-plate (refractive index 1.5) of thickness t is introduced in the path of one of the interfering beams (wave-length λ), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass-plate is
(A) 2λ (B) $2\lambda/3$
(C) $\lambda/3$ (D) λ

Space for rough work

Straight Objective Type

This section contains **5 multiple** choice questions numbered 11 to 15. Each question has 4 choices (A), (B), (C) and (D), out of which ***ONE or MORE*** are correct.

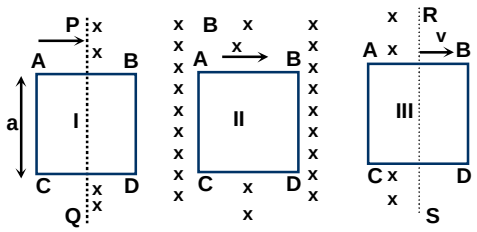
11. White light is used to illuminate the two slits in a Young's double slit experiment. The separation between the slits is b and the screen is at a distance D(>> b) from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing. Some of these missing wavelengths are

(A) $\lambda = \frac{b^2}{D}$ (B) $\lambda = \frac{2b^2}{D}$

(C) $\lambda = \frac{b^2}{3D}$ (D) $\lambda = \frac{b^2}{5D}$

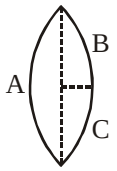
12. A square loop ABCD of edge 'a' moves to the right with a velocity v, parallel to AC. There is a uniform magnetic field of magnitude B, directed into the paper, in the region between PQ and RS only. I, II and III are three positions of the loop. Then,

(A) the current induced in the loop has magnitude Bav in all three positions
(B) the induced current is zero in position II
(C) the induced current is anticlockwise in position I
(D) the induced current is clockwise in position III



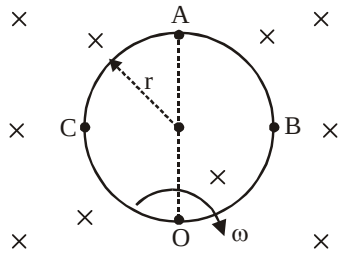
13. A thin equi-convex lens having power P is cut into three parts A, B and C as shown. The power of

(A) A is twice of B (B) B is equal to A
(C) C is equal to B (D) A is equal to C



Space for rough work

14. In figure, there is a conducting ring having resistance R placed in the plane of paper in a uniform magnetic field B_0. If the ring is rotating about in the plane of paper about an axis passing through point O and perpendicular to the plane of paper with constant angular speed ω in clockwise direction.



(A) Point A will be at higher potential than O.
(B) The potential of point B and C will be same.
(C) The current in ring will be zero.
(D) The current in the ring will be $\frac{2B_0\omega.r^2}{R}$.

15. X ray from a tube with a target A of atomic number Z shows strong K lines for target A and weak K lines for impurities. The wavelength of K_α lines is λ_z for target A and λ_1 and λ_2 for two impurities. $\frac{\lambda_z}{\lambda_1} = 4$ and $\frac{\lambda_z}{\lambda_2} = \frac{1}{4}$.

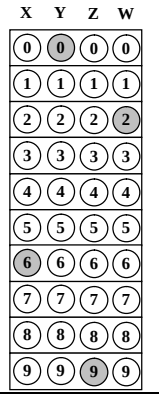
Screening constant of K_α lines to be unity. Select the correct statement(s)
(A) The atomic number of first impurity is 2z – 1.
(B) The atomic number of first impurity is 2z + 1.
(C) The atomic number of second impurity is $\frac{(z+1)}{2}$.
(D) The atomic number of second impurity is $\frac{z}{2}+1$.

Space for rough work

SECTION – C

Integer Answer Type

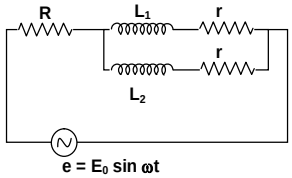
This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown



1. A plate of thickness t made of a material of refractive index $\mu = 2$ is placed in front of one of the slits in a double slit experiment. What should be the minimum thickness t in mm which will make the intensity at the centre of fringe pattern zero? ($\lambda = 4$ mm).

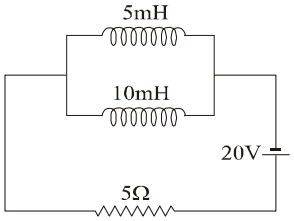
2. A bird in air looks at a fish vertically below inside water in a tank. The actual distance of bird and fish from the water level is equal but the distance of the fish as estimated by bird is S_1 and that of bird estimated by fish is S_2. [Given $(S_1/S_2) = 0.25$] then value of μ is _____ .

3. For the circuit shown in the figure, the rms value of voltage across R and coils are E_1 and E_2 respectively. The power developed across the coils in watts is K Watt. Find the value of K.
(Given $E_0 = 5\sqrt{2}$ V, $\omega = 100\pi$, $L_1 = 5$ mH, L_2=10 mH, r = 2Ω and R=2.5 Ω)



4. Radiations of two photons of energies twice & five times the work function of a metal, are incident successively on the metal surface. The ratio of maximum velocities of photoelectrons emitted in the two case is given as 1/x. Then what is value of x?

5. In the given circuit, the current through 5mH indicator in steady state is x/3 amp. Calculate value of x. (Neglect the mutual inductance)



Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

*This section contains 10 multiple choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **only one** is correct.*

1. Before adding the reagents of group (III) of basic radicals, the solution is heated with some concentrated HNO_3 in order to
   (A) lower the pH
   (B) oxidize Cr^{3+} to $Cr_2O_7^{2-}$
   (C) oxidize Fe^{2+} to Fe^{3+}
   (D) increase $[NO_3^-]$

2. When heated above 916°C, iron changes its crystal structure from bcc to ccp structure without any change in the radius of atom. The ratio of density of the crystal before heating and after heating is:
   (A) 1.089
   (B) 0.918
   (C) 0.725
   (D) 1.231

3. The aqueous solution of each of the following salt is coloured except
   (A) $TiCl_4$
   (B) $FeCl_3$
   (C) $CuCl_2$
   (D) $MnCl_2$

4. If all Fe^{2+} ions are replaced by Sr^{4+} ions in 0.1 mole of $Fe_{0.9}O$, then number of vacancies created are
   (A) 0.035 N_A
   (B) 0.01 N_A
   (C) 0.07 N_A
   (D) 0.1 N_A
   (N_A is Avogadro number.)

5. In diamond, the lattice is fcc with C atoms occupying lattice points as well as some of the voids. Which of the following statement(s) is/are correct about the diamond structure?
   (I) In the diamond lattice, half of the tetrahedral voids are occupied.
   (II) Percentage space occupied (packing efficiency) is 74%.
   (III) In the diamond lattice all octahedral voids are occupied.
   (A) Only I
   (B) Only III
   (C) I, II and III
   (D) Only II

Space for rough work

6. Na_2CO_3 cannot be used in place of $(NH_4)_2CO_3$ for the precipitation of group V cations, because
(A) Na^+ interferes in the detection of group V (B) concentration of CO_3^{2-} is very low
(C) Na^+ will react with the acid radicals (D) Mg^{2+} will precipitate as $MgCO_3$

7. When $K_4[Fe(CN)_6]$ is added to $FeCl_3$, the complex formed is
(A) $Fe_3[Fe(CN)_6]_4$ (B) $K_2Fe[Fe(CN)_6]$
(C) $Fe_4[Fe(CN)_6]_3$ (D) $K_2Fe[Fe(CN)_6]_2$

8. When excess of $SnCl_2$ is added to a solution of $HgCl_2$, a white precipitate that turn grey, is obtained. The grey colour is due to the formation of
(A) Hg_2Cl_2 (B) $SnCl_4$
(C) Sn (D) Hg

9. For an ideal binary liquid solution with $P_A^0 > P_B^0$, which relation between χ_A (mole fraction of A in liquid phase) and Y_A (mole fraction of A in vapour phase) is correct? (χ_B & Y_B are the mole fractions of B in liquid and vapour phase respectively)
(A) $\frac{Y_A}{Y_B} < \frac{\chi_A}{\chi_B}$ (B) $\chi_A > \chi_B$
(C) $Y_A < Y_B$ (D) $\frac{Y_A}{Y_B} > \frac{\chi_A}{\chi_B}$

10. The ratio of van't Hoff factor, if PBr_5 and PCl_5 are assumed to be soluble in water and remains same as in solid state, is
(A) 2 (B) 1
(C) 1.5 (D) 2.5

Space for rough work

Multiple Correct Answers Type

*This section contains 5 multiple correct answer(s) type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE** is/are correct.*

11. 1 mole benzene ($p^0_{benzene} = 42$ mm) and 2 mole toluene ($p^0_{toluene} = 36$ mm) will have:
    (A) total vapour pressure 38 mm
    (B) mole fraction of vapours of benzene above liquid mixture is $\frac{7}{19}$
    (C) mole fraction of vapours of benzene above liquid mixture is $\frac{1}{3}$
    (D) mole fraction of vapours of toluene above liquid mixture is $\frac{8}{19}$

12. In the equation, $X + CN^- + H_2O + O_2 \longrightarrow [X(CN)_2]^- + 4OH^-$; 'X' can be
    (A) Au (B) Ag_2S
    (C) Sb (D) As

13. 4.5g of Al (atomic mass = 27 amu) is deposited at cathode from Al^{+3} solution by a certain quantity of charge. The correct statement/s is are:
    (A) The volume of H_2 produced at 0°C and 1atm pressure from H^+ ions in solution by the same quantity of charge will be 5.6 *l*.
    (B) The volume of H_2 produced at 0°C and 1atm pressure from H^+ ions in solution by the same quantity of charge will be 22.4 *l*.
    (C) The number of moles of Al deposited will be 0.166 moles.
    (D) The number of faraday used for reduction of 1 mole of Al^{+3} is $\frac{3}{18}$F.

14. Which of the following statements are true regarding tetrahedral voids in fcc lattice?
    (A) Tetrahedral voids formed by one corner spheres and three face spheres.
    (B) Two tetrahedral voids can exist on one diagonal body line in fcc cubic.
    (C) Tetrahedral voids are equal to number of spheres involved in body.
    (D) Volume of tetrahedral voids are half of the octahedral voids.

15. Which of the following is/are the mixed salts?
    (A) Rochelle's salt (sodium potassium tartrate) (B) Bleaching powder
    (C) Calcium hypochlorite (D) Mohr's salt

Space for rough work

SECTION – C

(Integer Answer Type)

*This section contains **5** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

1. Freezing point of 0.2 M KCN solution is –0.7°C. On adding 0.1 mole of $Hg(CN)_2$ to one litre of the 0.2 M KCN solution, the freezing point of the solution becomes –0.53°C due to the reaction $Hg(CN)_2 + mCN^- \rightarrow Hg(CN)_{m+2}^{m-}$. What is the value of m assuming molality = molarity?

2. In the electrolysis of KI, I_2 is formed at the anode by the reaction;

   $2I^- \longrightarrow I_2 + 2e^-$

   After the passage of current of 0.5 ampere for 9650 seconds, I_2 formed required 40 ml of 0.1 M $Na_2S_2O_3 \cdot 5H_2O$ solution in the reaction;

   $I_2 + 2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + 2I^-$

   What is the current efficiency?

3. What is the oxidation number of Fe in violet coloured complex $Na_4\left[Fe(CN)_5(NO)S\right]$?

4. An alloy of copper, silver and gold is found to have cubic lattice in which Cu atoms constitute CCP. If Ag atoms are located at the edge centres and Au atom is present at body centre, the alloy will have the formula $Cu_xAg_yAu_z$. What is the value of X?

5. If $E^0_{Fe^{2+}/Fe} = 5V$ and $E^0_{Fe^{3+}/Fe^{2+}} = 2V$ then $E^0_{Fe^{3+}/Fe}$ will be.

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

*This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct*

1. If $|\vec{a}| = a$ and $|\vec{b}| = b$, then $\left(\frac{\vec{a}}{a^2} - \frac{\vec{b}}{b^2}\right)^2 =$

   (A) $\left(\frac{\vec{a}+\vec{b}}{ab}\right)^2$ (B) $\frac{(\vec{a}-\vec{b})^2}{ab}$

   (C) $\left(\frac{\vec{a}-\vec{b}}{ab}\right)^2$ (D) $\frac{(\vec{a}+\vec{b})^2}{ab}$

2. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-zero vectors such that no two are collinear $(\vec{a}\times\vec{b})\times\vec{c} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$. If θ is the acute angle between the vectors $\vec{b}$ and $\vec{c}$, then $\sin\theta$ equals

   (A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{\sqrt{2}}{3}$

   (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

3. The length and foot of the perpendicular from the point (2, –1, 5) to the line $\frac{x-11}{10} = \frac{y+2}{-4} = \frac{z+8}{-11}$ are

   (A) $\sqrt{14}, (1,2,-3)$ (B) $\sqrt{14}, (1,-2,3)$

   (C) $\sqrt{14}, (1,2,3)$ (D) None of these

Space for rough work

4. A box contains 15 tickets numbered 1, 2, ... 15. Seven tickets are drawn at random one after the other with replacement. The probability that the greatest number on a drawn ticket is 9, is

(A) $\left(\frac{9}{10}\right)^6$ (B) $\left(\frac{8}{15}\right)^7$

(C) $\left(\frac{3}{5}\right)^7$ (D) None of these

5. Image point of (5, 4, 6) in the plane $x + y + 2z - 15 = 0$ is

(A) (3, 2, 2) (B) (2, 3, 2)
(C) (2, 2, 3) (D) (– 5, –4, –6)

6. Out of 21 tickets marked with numbers from 1 to 21, three are drawn at random. The chance that the numbers on them are in A.P., is

(A) 10/133 (B) 9/133
(C) 9/1330 (D) None of these

7. If $A = \begin{vmatrix} \sin(\theta+\alpha) & \cos(\theta+\alpha) & 1 \\ \sin(\theta+\beta) & \cos(\theta+\beta) & 1 \\ \sin(\theta+\gamma) & \cos(\theta+\gamma) & 1 \end{vmatrix}$, then

(A) $A = 0$ for all θ (B) A is an odd Function of θ
(C) $A = 0$ for $\theta = \alpha + \beta + \gamma$ (D) A is independent of θ

8. If $A = \begin{bmatrix} \cos^2\theta & \sin\theta\cos\theta \\ \sin\theta\cos\theta & \sin^2\theta \end{bmatrix}$, $B = \begin{bmatrix} \cos^2\varphi & \sin\varphi\cos\varphi \\ \sin\varphi\cos\varphi & \sin^2\varphi \end{bmatrix}$

and θ and φ differs by $\frac{\pi}{2}$, then $AB =$

(A) I (B) O
(C) – I (D) None of these

Space for rough work

9. The area bounded by the curves $y = 2-|x-1|$, $y = \sin x$; $x = 0$ and $x = 2$ is

(A) $1+2\cos^2 1$ (B) $2+\sin^2 1$

(C) $\frac{\pi}{2}$ (D) $1+\log 2$

10. The solution of the equation $x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$ is

(A) $x\sin\left(\frac{x}{y}\right)+c=0$ (B) $x\sin y + c = 0$

(C) $x\sin\left(\frac{y}{x}\right)=c$ (D) None of these

Multiple Correct Choice Type

*This section contains **5 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct*

11. If A and B are two events such that $P(A) = 1/2$ and $P(B) = 2/3$, then

(A) $P(A \cup B) \geq 2/3$ (B) $P(A \cap B') \leq 1/3$

(C) $1/6 \leq P(A \cap B) \leq 1/2$ (D) $1/6 \leq P(A' \cap B) \leq 1/2$

12. The possible values of 'K' for which $\begin{bmatrix} \cos\frac{2\pi}{7} & -\sin\frac{2\pi}{7} \\ \sin\frac{2\pi}{7} & \cos\frac{2\pi}{7} \end{bmatrix}^K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is/are

(A) 7 (B) 6

(C) 8 (D) 9

Space for rough work

13. If a, b, c are distinct and $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix} = 0$

Then which of the following may be true

(A) a, b, c are in A.P. (B) a, b, c are in G.P.

(C) α is a root of $ax^2 - bx - c = 0$ (D) α is a root of $ax^2 + 2bx + c = 0$

14. If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and Q = P AP' then P'(Q^{2005}) P' is equal to

(A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} \frac{\sqrt{3}}{2} & 2005 \\ 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & -\frac{1}{2} \\ \frac{\sqrt{3}}{2} & 2005 \end{bmatrix}$ (D) $\begin{bmatrix} \frac{\sqrt{3}}{2} & -1 \\ 1 & 2005 \end{bmatrix}$.

15. If a, b, c are positive and distinct then which of the following cannot be roots of

$\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0$

(A) a (B) 0

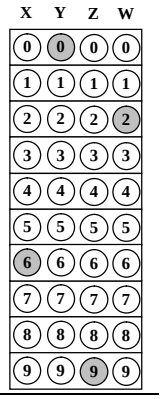
(C) b (D) c

Space for rough work

SECTION – C

Integer Answer Type

*This section contains **5 questions**. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown*



1. Let a, b > 0 and $\vec{\alpha} = \frac{\hat{i}}{a} + \frac{4\hat{j}}{b} + bk$ and $\vec{\beta} = b\hat{i} + a\hat{j} + \frac{1}{b}\hat{k}$ then the maximum value of $\frac{10}{5 + \vec{\alpha}.\vec{\beta}}$ is

2. The equation of the plane passing through the intersection of the planes $2x - 5y + z = 3$ and $x + y + 4z = 5$ and parallel to the plane $x + 3y + 6z = 1$ is x + 3y + 6z = k, where k is

3. A and B are two independent events. The probability that both A and B occur is 1/6 and the probability that at least one of them occurs is 2/3. Find 8P(A) + 9P(B) if P(A) > P(B).

4. The minimum area bounded by the function y= f(x) and y = αx + 9 ($\alpha \in R$) where f satisfies the relation f(x + y) = f(x) + f(y) + y$\sqrt{f(x)}$ $\forall x,y \in R$ and f′(0) = 0 is 9A, value of A is

5. If $x^3 \frac{dy}{dx} = y^3 + y^2\sqrt{y^2 - x^2}$ and $y = 1$ when $x = 1$, then $y = 2$ when $5x = ...(x \neq 0)$

Space for rough work